

Chapter 2: Sample Preparation for Analysis

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Contents

- Sample Preparation Techniques
- Steps in chemical analysis
- Preliminary treatment of sample

Learning Outcomes

- Explain the different techniques of sampling
- Describe sample preparation for analysis
- Determine the steps in chemical analysis

Introduction

- Sample preparation is the most complex development step in a pharmaceutical analysis.
 - The objective of sample preparation is to reproducibly provide 100% of the analyte in a solution that is ready for analysis.
- It is an essential part of the **separation process** and is one of the key considerations at the start of any method development strategy

Cont....

- Samples can be presented in various forms as solids, liquids, suspensions and even gases.
- When samples are in the solid form, they must first be either dissolved or extracted using a suitable solvent.
- In circumstances where the sample is relatively soluble, direct dissolution in a compatible solvent is the preferred mode of sample preparation because it is convenient and offers greater precision.

Introduction...

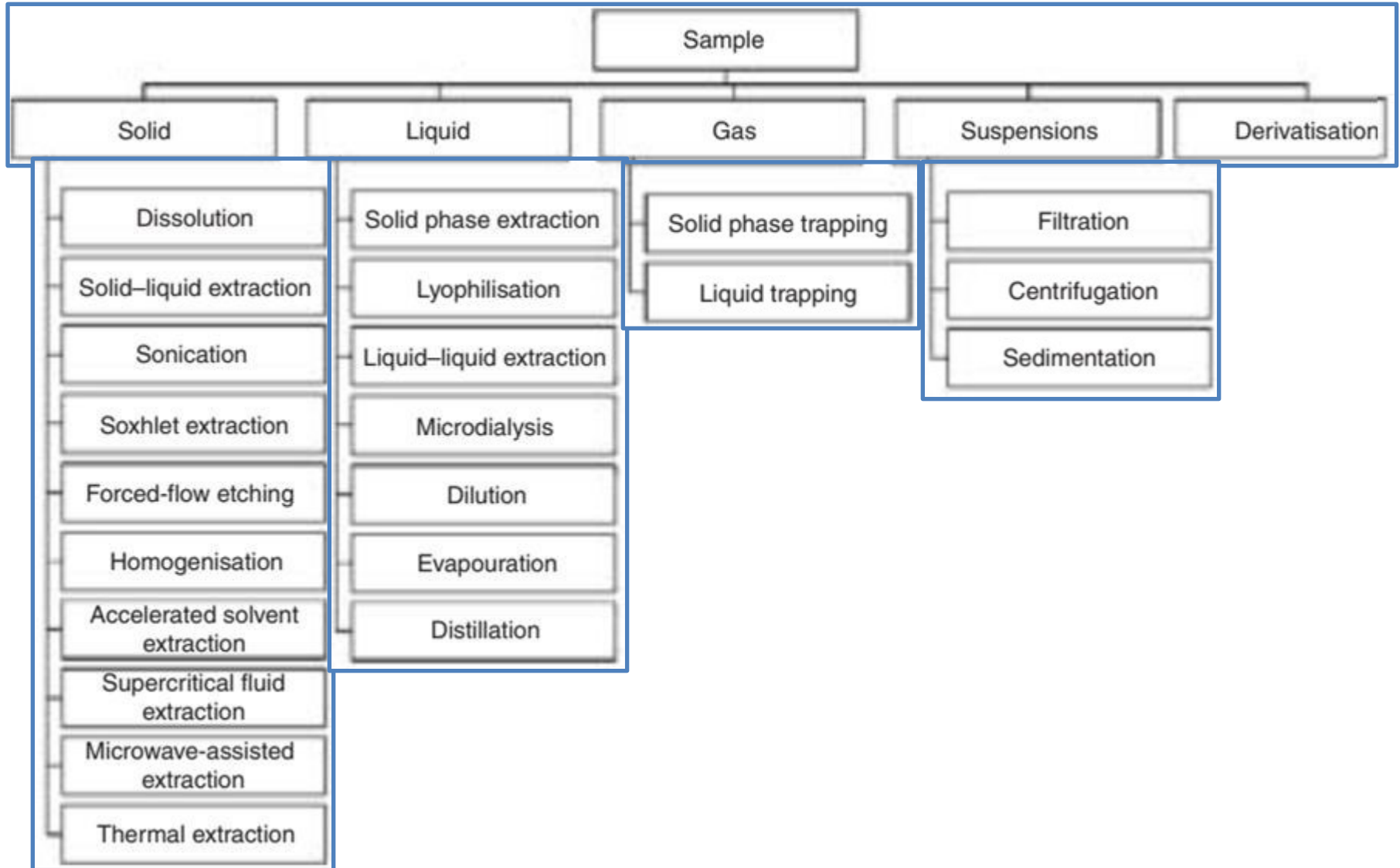


Fig. Diagram showing different modes of sample preparation methods currently available

Table: Summary of Sample Preparation Techniques

Technique	Application	Principles	Requirements
Liquid-liquid extraction	Isolation from liquid sample	Partitioning	Commonly available equipment and reagents
Solid phase extraction	Isolation from liquid sample	Step-gradient liquid chromatography mechanisms	Supplies are commonly available, including automated devices
Column-switching	In-line biological prep	Combines various liquid chromatography selectivities	Complex computer-controlled valve system
Solid-phase microextraction	Isolation from liquid samples, direct immersion or headspace	Partition or adsorption from solution or gas	Inexpensive segments of coated fused silica
Protein precipitation	Protein removal for plasma sample analysis	Decreases protein solubility using chaotropic agents	Commonly available equipment and reagents
Ultrafiltration	Protein removal for plasma sample analysis	Molecular-weight-selective membranes	Uses special filtration and low-speed centrifuge
Dialysis	Protein removal for plasma sample analysis	Molecular-weight-selective membranes combined with osmotic pressure	Dialysis membranes
Drug conjugates hydrolysis	Chemical hydrolysis of drug metabolites	Hydrolysis	Commonly available equipment and reagents
Direct HPLC injection	Analysis of plasma samples with no sample preparation	Achieves molecular-weight selectivity using specially designed bonded silica or maintains protein solubility with surfactants	Some methods require expensive columns
Derivatization	Chemical reaction to improve separation or detection	Treatment of active hydrogen functional group with electrophilic reagent	Small volumes of reactive derivatization reagents

Specific Sample Preparation Techniques

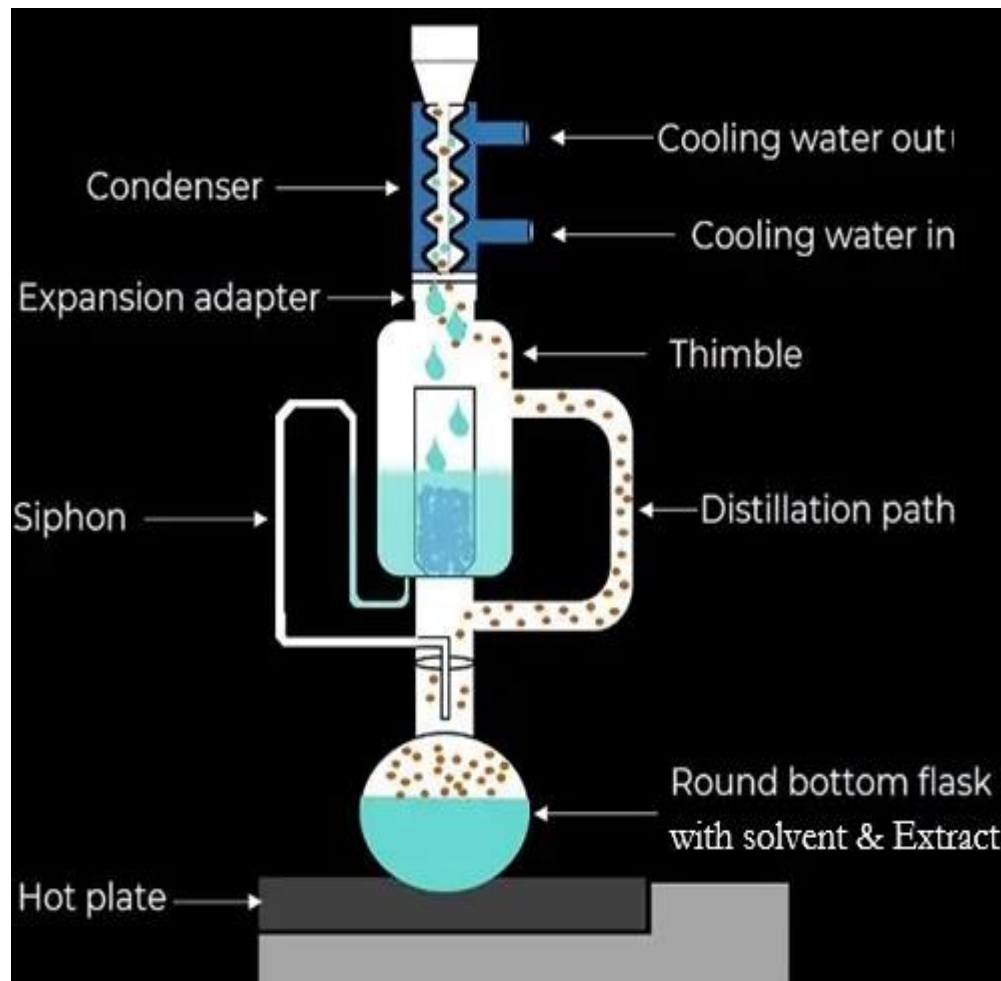
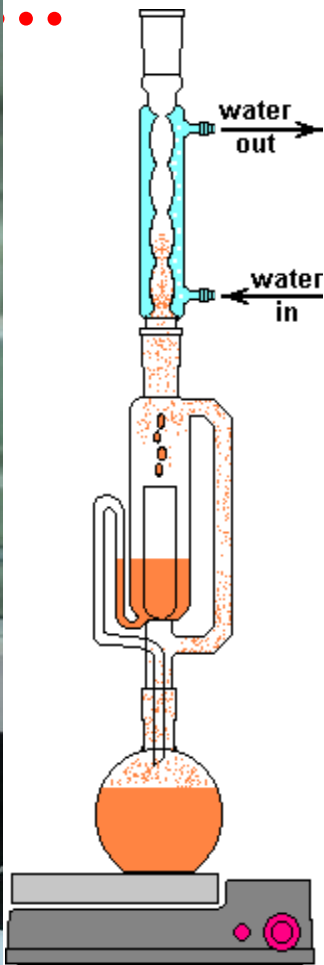
A. Soxhlet Extraction

- It is a **classical liquid-solid** sample preparation technique.
- The apparatus consists of a pot of extracting solvent which is refluxed to provide fresh hot solvent to drip through the solid sample.
- Solubility is maximized by using the clean hot solvent for dissolution.

- The sample is contained in a porous thimble held between the solvent flask and the reflux condenser.
- The Soxhlet technique is not often used in routine pharmaceutical sample preparations for two reasons:
 - ✓ The high temp. can cause degradation which is unacceptable for stability-indicating assays, and
 - ✓ The technique is difficult to perform on multiple samples because of time and space requirements.

Specific Sample Preparation Techniques

A. Soxhlet Extraction



Specific Sample Preparation Techniques...

B. Liquid–Liquid Extraction (LLE) (also called solvent extraction)

- It is generally reserved for more complex samples because it offers **poorer precision** than other techniques.
- It has been utilized in numerous applications including **petroleum, nuclear industries, hydrometallurgical, and pharmaceuticals**
- It is the **most commonly used for the preparation of biological samples** in which less precise methods can be tolerated.
- Occasionally an extraction is necessary for the determination of a **water-insoluble compound** in a **water-soluble matrix**, such as the analysis of **fat-soluble vitamins** in tablets or menthol in pharmaceutical lozenges.
 - To overcome poor extraction efficiencies, organic fraction from sequential extractions can be pooled

- Liquid extractions are also used for trace enrichment of the analyte by evaporating the organic layer to dryness and then reconstituting into a smaller volume of mobile phase.
- The addition of specific enhancers has been used to effect liquid–liquid extractions.

E.g. the extraction of phenobarbitol is improved by using specific binding compounds to enhance the **partition coefficient**.

Specific Sample Preparation Techniques...

B. Liquid–Liquid Extraction...

- **Limitation**
 - They are **very difficult to automate**.
- However, two approaches have been reported:
 - The automated LLE of **hydrochlorothiazide** from plasma & urine by using robotic system
 - Solvent extraction-flow injection, which is a different automation approach was originally described by Karlberg and Thelander
 - This is a continuous flow technique in which the aqueous sample is injected as segments into a stream of organic solvent.
 - Applied to the assay of **dimenhydrinate** with a precision of 2.5–4% and a throughput of 12 samples per hour.

Specific Sample Preparation Techniques...

C. Solid-Phase Extraction (SPE)

- It is a sample preparation technique that has been derived from liquid chromatography technology
- Applied extensively to the analysis of **biological samples** as well as **pharmaceutical products**.
- It is a step gradient technique in which the analyte, dissolved in a weak solvent, is retained on a stationary phase and subsequent additions of various moving phases of increasing solvent strength results in selective and controlled elution of the interferences and analytes.

There are **4 major steps** of solid phase extraction process:

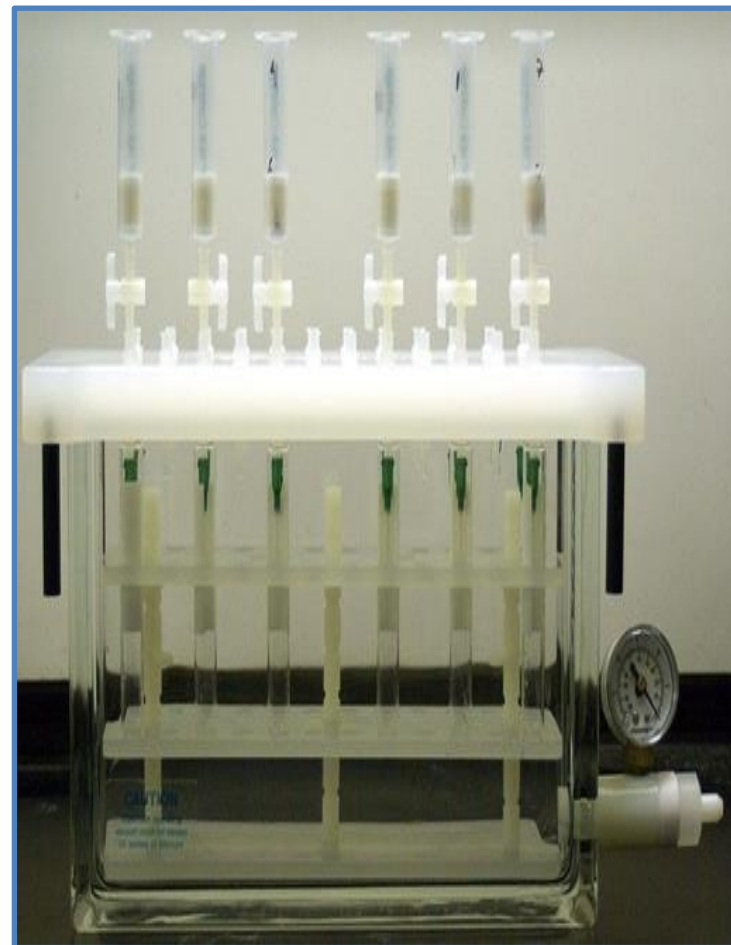
Conditioning, Sample loading, Washing, and Elution

- The final eluate is then analyzed directly or exposed to further treatment such as evaporation to concentrate the analyte or derivatization.

Specific Sample Preparation Techniques...

C. Solid-Phase Extraction (SPE) Apparatus

- A significant advantage that SPE offers over LLE is:
 - The **potential for automation of a routine procedure**
 - Generally **uses smaller volumes of organic solvents**
 - The **cost can be limited by reusing the columns** (extraction efficiency decreases only after two or three reuses)



Specific Sample Preparation Techniques...

D. Column-Switching Techniques

- It is applied extensively to the **HPLC analysis of drugs in biological fluids and other trace analysis in complex matrices**
 - Allowing the analyst to inject the sample directly, without prior sample preparation.
- According to Koenigbauer and Majors, the technique has many variations, which is classified into four types:
 - **Direct transfer, indirect transfer, reversed transfer, and loop transfer.**
- The analyte fraction of the first column eluate can be
 - Diverted directly to a second HPLC column (**direct transfer**), or
 - Eluted from column 1 to column 2 with a step gradient to a stronger mobile phase (**indirect transfer**), or
 - Eluted into a **loop injector** prior to the second column (**loop transfer**).
 - The combination of the two columns provides sufficient clean-up for a specific analysis
- **Reversed transfer:** requires a weak mobile phase in column 1 so that the analyte remains at the head of column 1.
- **Indirect transfer:** the stronger eluting mobile phase for column 1 becomes the separating mobile phase for column 2.
- **Direct transfer:** requires that the same mobile phase be used in columns 1 & 2

Specific Sample Preparation Techniques...

E. Solid-Phase Microextraction (SPME)

- It is applied to **trace analysis methods** such as the
 - **Analysis of flavour components, residual solvents, pesticides, leaching packaging components, or any other volatile organic compounds.**
- It is limited to **gas chromatography methods** because the sample must be desorbed by thermal means.
 - The mixture is quantitatively or qualitatively analyzed by GC.
- This preparation technique allows for **selective & solventless GC injections.**
- It is controlled by diffusion rates and partition effects

Specific Sample Preparation Techniques...

F. Protein Precipitation Methods

- HPLC analysis of plasma samples requires the removal of plasma protein.
- Solvent & solid-phase extraction methods achieve protein removal
- Proteins can also be **removed by precipitation & centrifugation** by adding reagents that **reduce the solubility of the proteins**.
 - Typical additives are **acetonitrile, methanol, or perchloric acid**.
- The resulting solution is then injected directly into the HPLC system
- The protein precipitation must also assure that the **drug is released by the protein**.
 - Many drugs, particularly **hydrophobic drugs**, are bound by plasma albumin protein by as much as **99% or higher**.

Specific Sample Preparation Techniques...

G. Sample Preparation of Drug Conjugates

- Compounds with poor water solubility are metabolized in humans to more **water-soluble compounds** to improve excretion rates.
 - **Acetaminophen**, for example, is metabolized to form glucuronide, sulfate, cysteine, and mercapturate conjugates
- **Full characterization of a drug's metabolism** requires analysis of both bound & free drug, particularly in urine, requiring analyses that are performed after **hydrolysis of the conjugates**.
- The **drug conjugate** can be hydrolyzed by the use of **acid, base, or enzyme catalysts**

Specific Sample Preparation Techniques...

H. Derivatization Techniques

- Chemical derivatization reactions have been used to **enhance detection, improve volatility for GC, or enhance selectivity** such as for **chiral separations**.
- Sample preparation derivatization reactions occur at an **active hydrogen** such as an **alcohol, phenol, amine, or sulfhydryl**. – The reaction types are largely **alkylations, acylations, silylations, & condensations**
- Many of the derivatization applications are developed to **improve a compound's volatility** so that gas chromatography (GC) may be employed.
 - The **presence of highly polar functional groups**, such as **carboxylic acids**, generally results in **poor volatility** for GC due to **intermolecular hydrogen bonding**.

Specific Sample Preparation Techniques...

H. Derivatization Techniques...

- Derivatization techniques are used to **mask these functional groups, improving volatility and allowing analysis by GC.**
- Derivatization reactions can improve **detection limits** in trace analysis by addition of **halogenated acyl or silyl groups** for subsequent analysis by GC with **electron-capture detection** or by addition of **a good UV chromophore or fluorophore** for subsequent analysis by HPLC
- **Selectivity enhancement for chiral analysis** can be done by derivatization of **racemic mixtures with a chiral agent.**

Introduction to Chemical Analysis

Introduction

- Chemical analysis is concerned with determining either the **identity of the chemical substances** or the **amount of a particular substance in a sample**.
 - The former type of analysis is a qualitative chemical analysis
 - The latter type is a quantitative chemical analysis
- Chemical analysis includes **any aspect of the chemical characterization of a sample material**
- Analytical Science** provides the methods and tools needed for insight into our material **world...for answering four basic questions about a material sample?**
 - What?
 - Where?
 - How much?
 - What arrangement, structure or form?

Types of Chemical

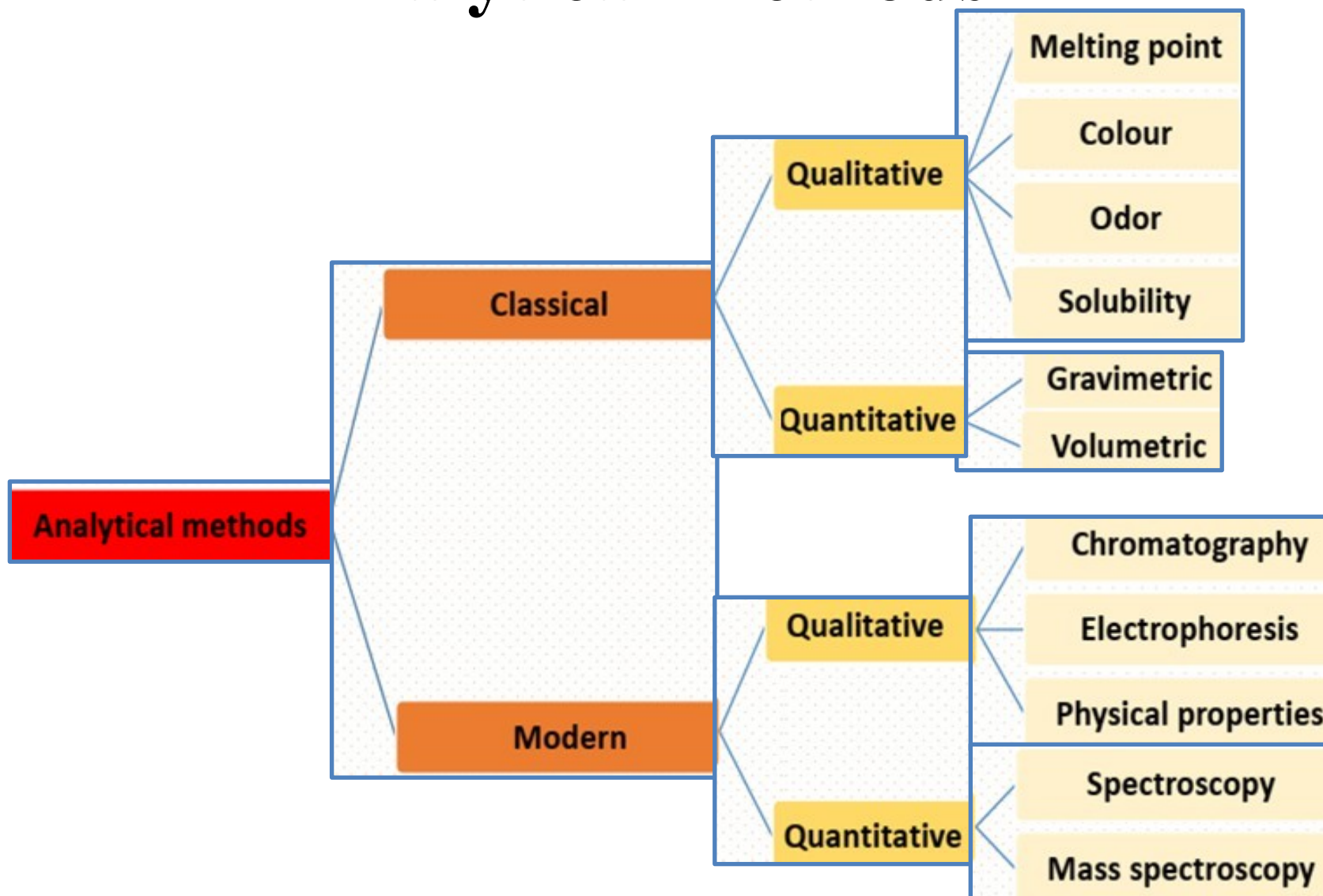
- **Four basic types** with respect to the information which is furnished
 - **Proximate Analysis:**
 - The amount of each component in a sample is determined with no concern as to the actual compounds present
 - **Partial analysis:**
 - Determination of selected components in a sample
 - **Trace constituent Analysis:**
 - Determination of specified components present in very minute quantity
 - **Complete Analysis:**
 - Proportion of each component of the sample is determined

Areas of Chemical

Questions they Answer:

- **Quantitation:**
 - How much of substance X is in the sample?
- **Detection:**
 - Does the sample contain substance X?
- **Identification:**
 - What is the identity of the substance in the sample?
- **Separation:**
 - How can the species of interest be separated from the sample matrix for better quantitation and identification?

Analytical Methods



Steps in Chemical Analysis

Steps in

- An analysis involves several steps and operations which depend on:
 - The particular problem, your expertise, the apparatus or equipment available.

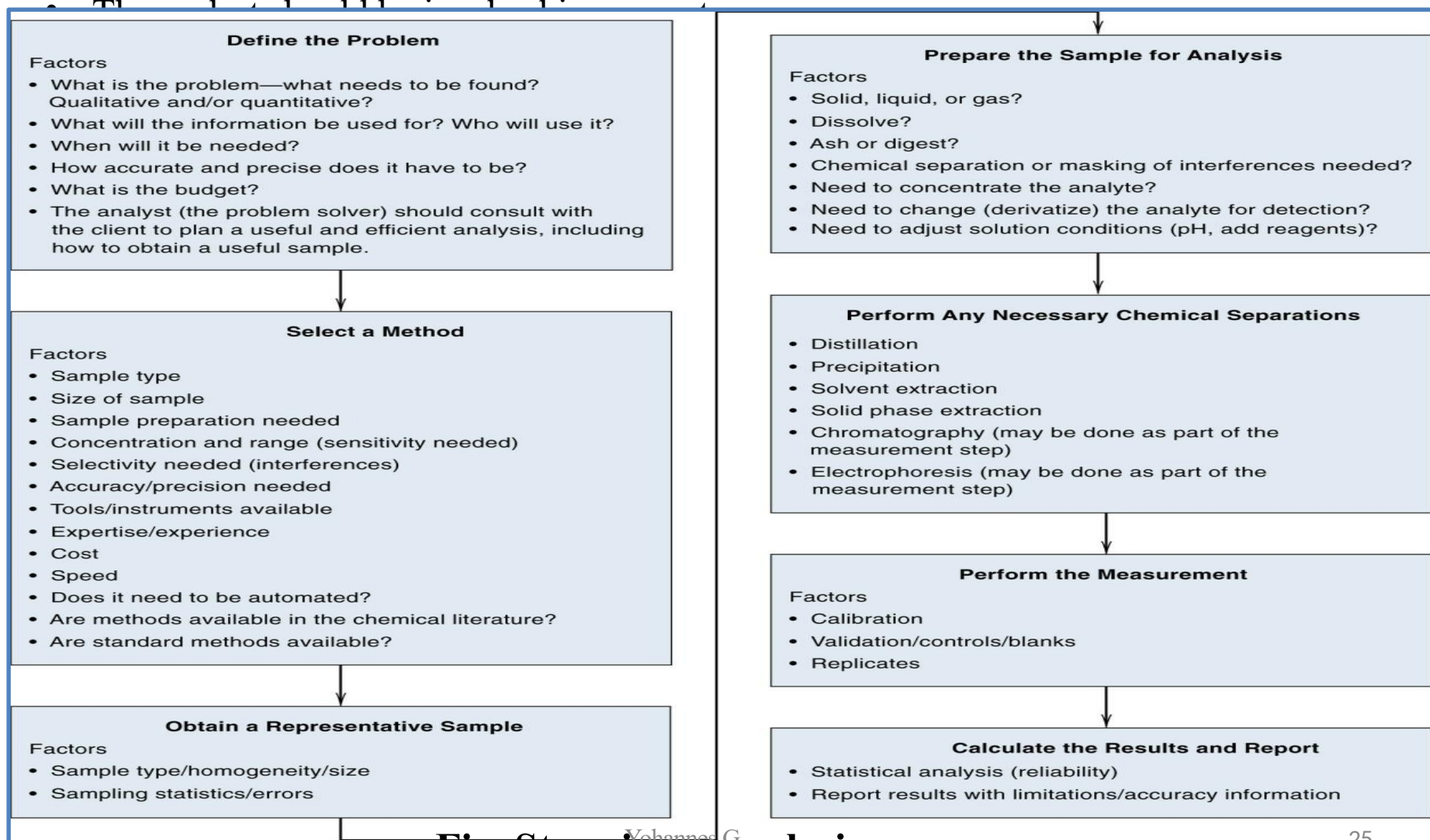


Fig. Steps in an analysis

Steps in an Analysis

Define the Problem

- **Factors**

- What is the problem? What needs to be found?
 - **Qualitative and/or Quantitative**
- What will the information be used for?
- Who will use it?
- When will it be needed?
- How accurate and precise should it be?
- The analyst (problem solver) should counsel the client to plan a useful and efficient analysis including how to find a useful sample

Steps in an Analysis...

Selection of a Method:

- Factors need to be considered in the selection process are:
 - Sample type and size required
 - Accuracy required
 - Sample preparation needed
 - Cost of analysis (total investment)
 - Number of sample to be analyzed
 - Time required for analysis
 - Skill required
 - Complexity of the sample
 - Number of components in the sample

Steps in an

Acquiring the Sample:

- Sampling is the **process of collecting a small mass of a material whose composition accurately represents the bulk** of the material from which it was taken.
- To **produce meaningful information**, an analysis must be performed on a representative sample.
- **For large heterogeneous sample**, great effort is required to get a **representative sample**.
- Sometimes **proper sampling is the most difficult step in an analysis and the source of greatest error**.
- The final result of an analysis will never be any more **reliable than the reliability of the sampling step**.
- **Types of sampling techniques**
 - Random sampling
 - Systematic sampling
 - Stratified sampling
 - Convenience sampling

Steps in an

Processing the Sample:

- Some times **no sample processing** is required prior to the measurement step
 - E.g. pH of water sample can be measured directly.
- Under most circumstances, sample need to be processed in a variety of different ways

i. Preparing a Laboratory Sample

ii. Defining Replicate Samples

iii. Preparing Solutions: Physical and Chemical Changes

Eliminating Interferences:

- Need to **eliminate substances from the sample** that may interfere with the measurement step.
- Few chemical or physical properties of importance in chemical analysis are **unique to a single chemical species**.
- **Species other than the analyte that affect the final measurement** are called interferences.
- **An interference causes an error in an analysis by enhancing or attenuating the quantity being measured.**
- **A scheme** must be devised to isolate the analytes from interferences before the final measurement is made.

Steps in an

Calibration and Measurement:

- An analytical results depend on a **final measurement of a physical or chemical property (X)**.
- Ideally, the measurement of the property is directly proportional to the **concentration (C_A)**.

$C_A = kX$, where, k is a proportionality constant.

- The process of determining k is an important step in most analyses.
 - This step is called a **calibration**.

Calculating Results:

- Analyte concentrations from experimental data need to be calculated.
- These computation are based on
 - The raw experimental data collected in the measurement steps
 - The characteristics of the measurement instruments and
 - The stoichiometry of the analytical reaction

Steps in an Analysis...

Evaluating Results by Estimating their Reliability:

- Analytical results are **incomplete without an estimate of their reliability.**
- **Some measure of the uncertainties** associated with computed results need to be provided.

Preliminary Treatment of Sample

Why do samples often need pre-treatment before analysis?

- Further processing the sample to make it suitable for analysis
- There are **three key issues** that often necessitate pre-treatment of a sample prior to analysis:
 - **The sample is in the wrong physical state** for the analysis method (e.g., the method requires a liquid but you have a solid sample),
 - **The sample has interfering matrix components** that may give either a false positive or negative reading in the measurement, and
 - **The sample has too low analyte concentration** to be detected by the instrument.

Challenges in Sample Preparation for Pharmaceutical Analysis

Complexity of Biological Matrices:

- Biological samples, such as blood or tissue, often contain a wide variety of components that can interfere with the analysis.
- complexity makes it difficult to isolate the target analytes without losing relevant information or introducing biases.

Variability in Techniques:

- Small differences in sample preparation techniques such as weighing, dilution, and mixing can lead to significant variability in results.
- variability can affect reproducibility and the overall reliability of the analysis.

Loss of Analytes:

- During the sample preparation process, there is a risk of losing target analytes due to adsorption to containers or degradation.
- It is particularly problematic for sensitive compounds that may degrade under certain conditions.

Interference from Excipients:

- In pharmaceutical formulations, excipients can interfere with the analysis of active ingredients.
- interference can complicate the interpretation of results and necessitate additional steps to mitigate its effects .

Robustness of Methods:

- Many sample preparation methods lack robustness, meaning they can be easily affected by slight changes in conditions or handling.
- It can lead to inconsistent results, which is particularly concerning in regulatory environments.

Time-Consuming Processes:

- Sample preparation can be labor-intensive and time-consuming, often considered the bottleneck in the analytical workflow.
- This can delay the overall analysis and increase costs.

***Thank
You***